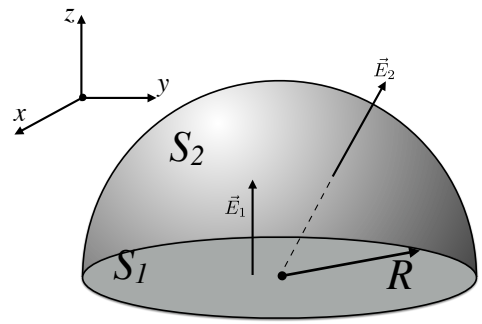
Problem 1 (50 points)

An unknown charge configuration resides inside a hemispherical closed surface with radius $R = 1$ m. The electric field is given by $\vec{E_1} = 10\ \hat{z}$ N/C on the circular surface S_1, and $\vec{E_2} = 5$ N/C (radially outward) on the bowllike surface S_2. (Hint: The direction of the area vector is chosen to be outward for a closed surface.)



(a) (15pts.) Find the electric flux Φ_E^1 on surface S_1.

$$\Phi_E^1 = \int \vec{E} \cdot d\vec{A} = -(10\ \text{N/C})\pi(1\ \text{m})^2 = -10\pi\ \text{N} \cdot \text{m}^2/\text{C}$$

(b) (15pts.) Find the electric flux Φ_E^2 on surface S_2.

$$\Phi_E^2 = \int \vec{E} \cdot d\vec{A} = (5\ \text{N/C})2\pi(1\ \text{m})^2 = 10\pi\ \text{N} \cdot \text{m}^2/\text{C}$$

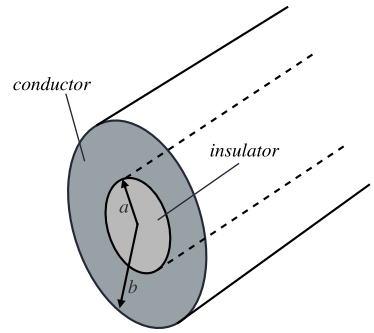
(c) (10pts.) Find the total flux Φ_E^T on the closed surface.

$$\Phi_E^T = \Phi_E^1 + \Phi_E^2 = 0$$

(d) (10pts.) Using Gauss' law, find the total charge inside the closed surface.

$$\Phi_E^T = q_{\text{encl}}/\epsilon_0 = 0 \longrightarrow q_{\text{encl}} = 0$$

Problem 2 (50 points)



A very long *conducting* cylindrical shell, with an inner radius a and an outer radius b, has a line charge density of $+2\lambda$. The cavity inside the shell is filled with an *insulating* material that carries a line charge density of $-\lambda$ distributed uniformly through its volume.

(a) (25pts.) Find the magnitude of the electric field as a function of r (the radial distance from the center of the cylinders) for the following regions: i) $\boldsymbol{r \leq a}$ ii) $\boldsymbol{a \leq r \leq b}$ iii) $\boldsymbol{b \leq r}$

i) $\boldsymbol{r \leq a}$

$$E(2\pi r\ell) = q_{\text{enc}}/\epsilon_0 = \frac{q}{(\pi a^2 \ell)}(\pi r^2 \ell)$$

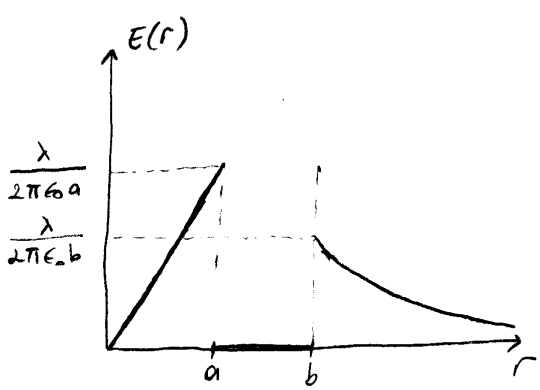
$$E = \frac{1}{2\pi\epsilon_0}\frac{q}{\ell}\frac{r}{a^2} \longrightarrow \boxed{E(r) = \frac{1}{2\pi\epsilon_0}\lambda\frac{r}{a^2}}$$

ii) $\boldsymbol{a \leq r \leq b}$ $\boxed{\text{E=0}}$ (inside conductor)

iii) $\boldsymbol{b \leq r}$

$$E(2\pi r\ell) = q/\epsilon_0 \longrightarrow \boxed{E(r) = \frac{1}{2\pi\epsilon_0}\frac{\lambda}{r}}$$

(b) (10pts.) Plot $E(r)$ as a function of r.



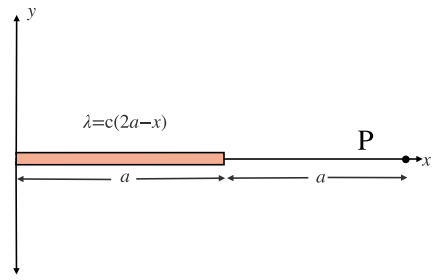
(c) (15pts.) Find the line charge densities on the inner and outer surfaces of the conducting cylindrical shell.

inner surface: $+\lambda$

outer surface: $+\lambda$

Problem 3 (50 points)

Positive charge Q is distributed along a rod lying on the x-axis from $x = 0$ to $x = a$ with the line charge density $\lambda = c\,(2a - x)$, where c is a constant.



(a) (15pts.) Find c in terms of the total charge Q and the length of the rod, a.

$$Q = \int_0^a \lambda(x)dx = c\int_0^a (2a - x)\,dx = c(2ax - x^2/2)\Big|_0^a = c(3a^2/2) \longrightarrow \boxed{c = \frac{2Q}{3a^2}}$$

(b) (35pts.) Find the magnitude of the electric field produced by the charge distribution on the positive x-axis at $x = 2a$, a distance a to the right of the end of the rod. (Express your answer in terms of Q and a.)

$$E = \int_0^a \frac{1}{4\pi\epsilon_0}\frac{dq}{(2a - x)^2}$$

$$dq = \lambda(x)dx = c(2a - x)dx$$

$$E = \frac{c}{4\pi\epsilon_0}\int_0^a \frac{dx}{2a - x} = -\frac{c}{4\pi\epsilon_0}\ln(2a - x)\Big|_0^a = \frac{c}{4\pi\epsilon_0}\ln 2$$

$$\boxed{E = \frac{Q}{6\pi\epsilon_0}\frac{\ln 2}{a^2}}$$